

FEA Assessment – Humanoid Leg (3D Printed)

Overview

You are provided with a CAD assembly of a humanoid robot leg. Your task is to perform a finite element analysis (FEA) to evaluate its structural performance under realistic loading conditions.

This assessment focuses on your **engineering approach, assumptions, and interpretation**, not just software usage.

Instructions

- Focus on **quality over speed**
 - Clearly justify all assumptions and decisions
 - Use **ANSYS** for all simulations (exclusive requirement)
-

Problem Context

The leg assembly is part of a humanoid robot and is assumed to be **3D printed**.

Additional Requirement (Comparison Study)

You must also perform a **comparison study**:

- Assume the same geometry made from **uniform isotropic aluminium**
- Compare results between:
 - 3D printed (anisotropic assumption)
 - Aluminium (isotropic assumption)

Comment on:

- Difference in stress distribution
- Difference in deformation
- Design implications

The leg assembly is part of a humanoid robot and is assumed to be **3D printed**.

Assume:

- Total robot mass = **80 kg**
- Worst-case condition = **entire weight on one leg**
- Gravity = **9.81 m/s²**

You may introduce additional assumptions if required, but they must be clearly justified.

Design & Manufacturing Consideration (3D Printing Workflow)

Since the leg is assumed to be **3D printed**, you are expected to consider how CAD translates to a 3d printed STL.

1. CAD to STL Preparation

- Export CAD geometry to STL format
- Ensure appropriate resolution (avoid overly coarse triangulation)
- Maintain geometric accuracy in critical regions

2. Part Orientation Strategy

- Decide optimal print orientation
- Balance between:
 - Strength (layer direction)
 - Support requirement
 - Surface finish

3. Assembly Strategy

- Determine if parts should be printed as a single unit or split
 - Consider ease of printing vs structural integrity
-

FEA Methodology (Specific to 3D Printed Parts)

1. Geometry Preparation

- Simplify unnecessary small features (fillets, threads if not critical)
- Ensure proper contact definitions between parts

2. Material Consideration (IMPORTANT)

3D printed parts are **anisotropic** (properties vary by direction).

- Define material properties carefully
- If exact data is unavailable, clearly state assumptions
- Consider print orientation (layer direction)

3. Print Orientation Awareness

- Identify likely print direction
- Consider weaker strength in layer bonding direction (Z-direction)
- Evaluate if critical stresses align with weak directions

4. Boundary Conditions

- Apply realistic constraints (avoid over-constraining)
- Ensure load path is physically meaningful

5. Meshing Strategy

- Use finer mesh near stress concentrations
- Ensure element quality is acceptable
- Avoid excessively coarse mesh in thin regions

6. Analysis Approach (Part vs Assembly)

a) Individual Part Analysis

- Analyze critical components separately
- Helps identify weak parts quickly
- Allows finer mesh and faster iteration

b) Assembly-Level Analysis

- Analyze full system with contacts/interactions
- Captures realistic load transfer between components

You are encouraged to:

- Start with simplified or individual part analysis
- Then validate using full assembly simulation

7. Validation Mindset

- Check if deformation pattern makes physical sense
 - Identify unrealistic stress spikes (possible singularities)
-

Task Requirements

1. Model Understanding

- Review the CAD assembly
 - Identify load-bearing components (links, joints)
 - Highlight critical regions
-

2. Assumptions

Clearly define:

- Material properties
 - Load distribution
 - Contact/joint behavior
-

Modeling Note

- Actuators should be **simplified and modeled as solid rigid blocks** (no internal details required)
-

3. Boundary Conditions & Loading

- Apply realistic ground constraint at foot
 - Apply body weight load
 - Ensure realistic load transfer through leg
-

4. Meshing

- Choose appropriate element types
 - Generate mesh
 - Refine near joints and stress areas
 - Justify mesh strategy
-

5. Analysis

Perform:

- Linear static structural analysis

Extract:

- Total deformation
 - Von Mises stress
 - Maximum stress locations
-

6. Interpretation

- Identify critical/high-stress regions
 - Evaluate safety of design
 - Comment on stress concentration and deformation
 - Highlight modeling limitations
-

7. Design Suggestions

Suggest at least **2 improvements**, such as:

- Improving strength at joints
 - Reducing weight
 - Improving load distribution
-

Deliverables

Submit a report (PDF or PPT) including:

1. Assumptions
 2. Boundary conditions and loads
 3. Mesh screenshots
 4. Result plots (stress, deformation)
 5. Key findings
 6. Design recommendations
-

Important Notes

- There is **no single correct answer**
 - Focus on **engineering reasoning over perfect results**
-

Critical Engineering Questions (Must Answer)

Include clear answers in your report:

- Would you actually manufacture this design? Why or why not?
 - What is the biggest risk in this design under real-world use?
 - Which component would you redesign first, and why?
 - How does 3D printing orientation affect your final design decision?
 - Is the aluminium version more practical than the 3D printed one? Justify.
-

Evaluation Focus

You will be evaluated on:

- Quality of assumptions
 - Correctness of setup
 - Understanding of results
 - Engineering judgment
 - Clarity of communication
-

End of Task